

SRS (Software Requirement Specification)

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What is an SRS ?

- SRS is the official statement of what the system developers should implement.
- SRS is a complete description of the behaviour of the system to be developed.
- SRS should include both a definition of user requirements and a specification of the system requirements.
- The SRS fully describes what the software will do and how it will be expected to perform.

Purpose of SRS

- The SRS precisely defines the software product that will be built.
- SRS used to know all the requirements for the software development and thus that will help in designing the software.
- It provides feedback to the customer.

SRS Format

Introduction

- Purpose
- Document Conventions
- Product Scope
- Reference

Overall Description

- Product Perspective
- Product Functions
- User Classes and Characteristics
- Operating Environment
- Design and Implementation Constraints
- User Documentation
- Assumptions and Dependencies

External Interface Requirements

- User Interfaces
- Hardware Interfaces
- Software Interfaces
- Communications Interfaces

Continue...

System Features

- System Feature 1
- System Feature 2
(and so on)

Other Nonfunctional Requirements

- Performance Requirements
- Safety Requirements
- Security Requirements
- Software Quality Attributes
- Business Rules

Other Requirements

- Appendix A: Glossary
- Appendix B: Analysis Models

Types of reader for requirement specification

System Customers

Specify the requirements and read them to check that they meet their needs. Customer specify changes to the requirements .

Managers

Use the Requirements Document to plane a bid for the system and to plan system development process.

System Engineers

Use the requirements to understand what system is to be deployed.

IEEE Standard for SRS

Source: Adapted from IEEE-STD-830-1993 See also, Blum 1992, p160

1 Introduction

Purpose

Scope

Definitions, acronyms, abbreviations

Reference documents

Overview

Identifies the product, & application domain

Describes contents and structure of the remainder of the SRS

Describes all external interfaces: system, user, hardware, software; also operations and site adaptation, and hardware constraints

2 Overall Description

Product perspective

Product functions

User characteristics

Constraints

Assumptions and Dependencies

Summary of major functions, e.g. use cases

Anything that will limit the developer's options (e.g. regulations, reliability, criticality, hardware limitations, parallelism, etc)

3 Specific Requirements

Appendices

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All the requirements go in here (i.e. this is the body of the document). IEEE STD provides 8 different templates for this section

Continue...

System test engineers

Use the requirements to develop validation tests for the system

System Maintains engineers

Use the requirements to understand the system and the relationship between its parts

Characteristics of SRS

- **Correct** : Every requirement given in SRS is a requirement of the software.
- **Unambiguous**: Every requirement has exactly one interpretation.
- **Complete**: Includes all functional, performance, design, external interface requirements; definition of the response of the software to all inputs.
- **Consistent**: Internal consistency.
- **Ranked importance**: Essential vs. desirable.

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- **Verifiable:** A requirement is verifiable if and only if there exists some finite cost effective process with which a person or machine can check that the SW meets the requirement.
- **Modifiable:** SRS must be structured to permit effective modifications (e.g. don't be redundant, keep requirements separate)
- **Traceable:** Origin of each requirement is clear.

what are the benefits of a great SRS?

The IEEE 830 standard defines the benefits of a good SRS:

- Establish the basis for agreement between the customers and the suppliers on what the software product is to do.
- Reduce the development effort.
- Provide a basis for estimating costs and schedules.
- Provide a baseline for validation and verification.
- Serve as a basis for enhancement.

Thank-You

